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Winners, Losers, and Regulators in a Derivatives Market Bubble

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1 Big picture

- **Question.** How is money made and lost inside a financial market bubble? The paper studies trader-level behavior in the Chinese warrants bubble.
- **Setting.** Chinese stock warrants are attractive for this question because their fundamental values are transparent under Black-Scholes, yet many traded far above those values for sustained periods.
- **Data advantage.** The authors use proprietary brokerage account data with complete trading histories and day-end holdings for a large investor base. This allows them to distinguish individuals from institutions and to track each trader's profits, turnover, and holdings.
- **Main empirical finding.** Most investors lose money, but a small group of persistently successful investors earns high returns. These winners are not simply lucky: they trade extremely actively and behave like de facto market makers.
- **Winner behavior.** Skilled traders buy when prices fall and sell when prices rise. Their profits come from liquidity provision and short trading cycles, not from holding the bubble asset until the end.
- **Loser behavior.** Unskilled investors trend-chase and increase positions in out-of-the-money warrants near expiration, even when warrants are essentially worthless.
- **Regulatory finding.** Chinese warrant price limits often failed to bracket fundamental values. Some lower price limits were above Black-Scholes values and even above maximum possible payoffs.
- **Economic takeaway.** Bubble losses arise from a combination of investor naivete, option-payoff misunderstanding, shorting/frictional constraints, and poorly calibrated regulatory rules. Sophisticated traders can profit by supplying immediacy to enthusiastic traders, but only a small minority does so persistently.

2 Data and measurement

2.1 The Chinese warrants bubble

- The sample covers the Chinese warrants market during 678 trading days from January 2007 to October 2009.
- The paper studies 48 stock warrants traded on the Shanghai and Shenzhen exchanges.
- A warrant has a clean benchmark value because it is a derivative claim on an underlying stock. The paper uses Black-Scholes estimates as the primary fundamental-value benchmark.
- The market was extremely active: daily turnover is often above 100%, and many investors traded intraday rather than holding to maturity.
- The market also included price limits, trading suspensions, eligibility rules, and risk disclosure requirements. These institutional details are central to the regulatory analysis.

Interpretation. The setting is useful because it combines a visible bubble with trader-level records and a relatively transparent valuation benchmark. This is rare in bubble research.

2.2 Investor-level data

- The brokerage data include account-level trades, timestamps, order directions, prices, quantities, and day-end holdings.
- The paper observes whether an account is an institution or an individual.
- The active analysis sample in the descriptive statistics contains 223,745 investors.

- The paper excludes investors who initially received endowed warrants when measuring trading performance, because endowed positions would distort profits and returns.
- Trading records are matched to warrant characteristics, Black-Scholes values, price limits, and market trading records.

Interpretation. The key advantage is that the paper can see who bought, who sold, when they traded, and whether they subsequently profited or lost. This permits a direct decomposition of bubble outcomes across trader types.

2.3 Daily rate of return: DRR

The paper constructs a daily rate of return for investor i on day t :

$$DRR_{it} = \frac{Trading_Revenue(i, t) + Holding_Value_Change(i, t)}{Maximum_Cash_Used(i, t) + Holding_Value(i, t - 1)}. \quad (1)$$

The holding-value change is

$$Holding_Value_Change(i, t) = Holding_Value(i, t) - Holding_Value(i, t - 1). \quad (2)$$

The maximum cash used is

$$Maximum_Cash_Used(i, t) = \max \left\{ 0, - \min_{1 \leq j \leq n_{it}} \sum_{k=1}^j CashFlow_{i,t,k} \right\}. \quad (3)$$

Interpretation. DRR is designed for a day-trading environment. A holdings-based return would miss much of the action because many investors trade warrants intraday without carrying positions overnight.

2.4 Daily turnover: DTO

The paper defines a cash-value-based daily turnover measure:

$$DTO_{it} = \frac{Turnover_Value(i, t)/2}{Maximum_Cash_Used(i, t) + Holding_Value(i, t - 1)}. \quad (4)$$

- DTO is not capped at 100%.
- This is important because warrant-market turnover is extremely high.
- A high DTO can indicate market-making-like behavior if it is paired with short holding durations and consistent profitability.

Interpretation. DTO is one of the main tools used to distinguish sophisticated liquidity provision from ordinary speculative trading.

2.5 Performance classification

- Investors are ranked using Sharpe ratios of their daily net DRRs.
- The paper uses a holdout period to identify ex ante top performers and bottom performers.
- Top performers are tested against bootstrap simulations to separate skill from luck.
- The most important subgroup is the top 1% of investors by ex ante Sharpe ratio, and especially the high-DTO subgroup within them.

Interpretation. The holdout-sample classification avoids the simplest form of look-ahead bias. It asks whether the investors who looked good early continued to look good later.

2.6 Regulatory variables

- The paper compares market prices, price limits, and Black-Scholes values.
- It identifies days when the lower price limit is above the warrant’s fundamental value.
- It also identifies cases where the price limit is above the maximum possible payoff of the warrant.
- These observations reveal whether regulatory price limits anticipated the bubble’s possible overvaluation.

Interpretation. A price limit can be stabilizing only if it brackets economically plausible values. If the lower limit is still above fundamental value, it can preserve overpricing rather than correct it.

3 Empirical specifications and tests

3.1 Investor-level performance regressions

The paper estimates panel regressions of the form

$$Y_{it} = \alpha_t + \beta Individual_i + \Gamma X_{it} + \varepsilon_{it}, \quad (5)$$

where Y_{it} is gross DRR, net DRR, or DTO; α_t are day fixed effects; and X_{it} includes option and investor controls such as skewness, age, gender, account age, relative value, leverage, and volatility.

Interpretation. The coefficients measure whether individuals perform or trade differently from institutions after absorbing market-wide daily conditions and option characteristics.

3.2 Luck-versus-skill bootstrap

For each investor, the paper simulates adjusted daily returns by resampling from the investor’s return distribution. It compares the actual t -statistic of mean DRR with the distribution from 10,000 simulations.

- A top investor is skillful if the actual t -statistic exceeds simulated values at very high percentiles.
- A bottom investor is persistently poor if the actual result is worse than what would be expected under the simulated benchmark.

Interpretation. This procedure asks whether observed top performance is too strong to be attributed to chance alone.

3.3 Event study around top traders’ transactions

The paper centers event time around top investors’ buy or sell trades and plots price changes and net buying by top and bottom investors.

Interpretation. If top investors are liquidity providers, they should buy after short-run price declines and sell after short-run price increases. If bottom investors are trend chasers, their trading should move in the opposite direction.

3.4 Supertrader event study

The paper identifies supertraders within the top investors as those with very high DTO. It then estimates returns around episodes in which supertraders have large net buys or large net sells.

Interpretation. Large supertrader buys occur after price drops, and large supertrader sells occur after price rises. This is the signature of market-making rather than simple directional speculation.

3.5 Out-of-the-money and zero-payoff tests

The paper studies OTM warrants near expiration and warrants with guaranteed zero payoff. It tracks holdings, trading volume, and investor groups in the final trading days.

Interpretation. These tests identify naivete directly. Buying or holding a warrant with zero payoff or very little remaining value is difficult to rationalize as informed speculation.

3.6 Price-limit tests

For each warrant, the paper counts trading days on which the lower price limit exceeds Black-Scholes values or maximum possible payoffs.

Interpretation. This test evaluates the regulator’s design problem. If regulatory limits are too high, they can prevent prices from falling to fundamentals and leave naive investors trapped.

4 Identification and empirical design

4.1 What makes the evidence credible

- The paper observes actual investor-level trades and holdings, not survey beliefs or aggregate order flow.
- The warrant setting has transparent valuation benchmarks, so overpricing can be identified more directly than in equity bubbles.
- Performance is ranked in a holdout period and evaluated later, which makes persistence evidence more convincing.
- The paper uses event-time evidence around trades, final maturity, price-limit hits, and trading suspensions to connect behavior to economically meaningful events.

Interpretation. The design is descriptive rather than a randomized experiment, but the granularity of the data and the transparent derivative benchmarks make the behavioral interpretations strong.

4.2 Main threats and responses

- **Luck versus skill:** addressed with bootstrap simulations and out-of-sample performance.
- **Risk adjustment:** daily warrant returns are difficult to risk-adjust conventionally, so the paper uses mean-adjusted return simulations and holdout-sample classification.
- **Institutional versus retail differences:** addressed by separating institutions and individuals and using investor-level controls.
- **Valuation uncertainty:** the use of warrants reduces but does not eliminate model-dependence in fundamentals. The paper therefore also uses maximum payoff comparisons for some regulatory tests.
- **Generalizability:** Chinese warrants were unusual, with high turnover and retail participation; the paper’s contribution is strongest for understanding bubbles in derivative-like assets with naive participation.

Interpretation. The empirical goal is to document who wins and loses inside a bubble and what mechanisms produce those outcomes. The study is less about estimating one universal causal coefficient and more about isolating detailed trader behavior.

5 Tables and figures

The visuals below are directly cropped from the original paper and grouped to save space.

5.1 Table 1 + Table 2: Warrants and market-wide investor outcomes

Read:

- Table 1 lists all 48 warrants, with type, underlying, share supply, moneyness, and trading periods.
- The trading periods show that many warrants expire in the 2007-2009 window, allowing the paper to examine last-day and maturity behavior.
- Table 2 shows total turnover of about 1.50 trillion yuan in the active sample.
- Aggregate gross profits are negative, and aggregate net profits are even more negative after transaction costs.
- Individuals lose much more than institutions: individuals account for almost the entire aggregate net loss.
- Average net DRR is negative and turnover is very high, with mean DTO around 82%.

Economic message. The bubble is not a small or thin-market anomaly. It is a high-turnover market in which most investors lose money, especially after trading costs.

SN	Name	Code	Type	Source	Share	MShare	Money	Trading period	
								Begin	End
1	WuLiang	030002	Call	1	298	—	1.666	04/03/2006	03/26/2008
2	QiaoCheng	031001	Call	1	145	—	2.156	11/24/2006	11/16/2007
3	GangFan	031002	Call	0	800	—	0.991	12/12/2006	12/04/2008
4	ShenFa	031003	Call	1	155	—	0.660	06/29/2007	12/21/2007
5	ShenFa	031004	Call	1	78	—	0.100	06/29/2007	06/20/2008
6	GuoAn	031005	Call	0	96	—	0.062	09/25/2007	09/17/2009
7	ZhongXing	031006	Call	0	65	—	0.007	02/22/2008	02/05/2010
8	EJiao	031007	Call	1	131	—	1.259	07/18/2008	07/10/2009
9	GangFan	038001	Put	1	233	—	-1.222	11/04/2005	04/24/2007
10	HuaLing	038003	Put	1	633	—	-0.970	03/02/2006	02/22/2008
11	WuLiang	038004	Put	1	313	—	-1.527	04/03/2006	03/26/2008
12	ZhongJi	038006	Put	1	424	—	-1.194	05/25/2006	11/16/2007
13	JiaFei	038008	Put	1	120	—	-1.097	06/30/2006	06/22/2007
14	BaoGang	580002	Call	1	715	919	1.078	03/31/2006	03/23/2007
15	HanGang	580003	Call	1	926	1,116	0.914	04/05/2006	03/28/2007
16	ShouChuang	580004	Call	1	60	259	0.818	04/24/2006	04/16/2007
17	WanHua	580005	Call	1	57	62	1.804	04/27/2006	04/19/2007
18	YaGe	580006	Call	1	91	253	1.977	05/22/2006	05/14/2007
19	ChangDian	580007	Call	1	407	—	0.996	05/25/2006	05/17/2007
20	GuoDian	580008	Call	1	151	255	1.162	09/05/2006	08/28/2007
21	YiLi	580009	Call	1	111	—	1.291	11/15/2006	11/07/2007
22	MaGang	580010	Call	0	1,265	—	0.229	11/29/2006	11/14/2008
23	ZhongHua	580011	Call	0	180	—	1.126	12/18/2006	12/10/2007
24	YunHua	580012	Call	0	54	—	0.441	03/08/2007	02/20/2009
25	WuGang	580013	Call	0	728	—	-0.260	04/17/2007	04/09/2009
26	ShenGao	580014	Call	0	108	—	-0.846	10/30/2007	10/22/2009
27	RiZhao	580015	Call	0	62	—	-0.322	12/03/2007	11/18/2008
28	ShangQi	580016	Call	0	227	—	-0.011	01/08/2008	12/30/2009
29	GanYue	580017	Call	0	56	—	-0.228	02/28/2008	02/05/2010
30	ZhongYuan	580018	Call	0	51	—	-0.498	02/26/2008	08/18/2009
31	ShiHua	580019	Call	0	3,030	—	-0.541	03/04/2008	02/24/2010
32	ShangGang	580020	Call	0	292	—	-0.774	03/07/2008	02/27/2009
33	QingPi	580021	Call	0	105	—	0.073	04/18/2008	10/12/2009
34	GuoDian	580022	Call	0	427	—	0.000	05/22/2008	05/14/2010
35	KangMei	580023	Call	0	167	—	0.432	05/26/2008	05/18/2009
36	BaoGang	580024	Call	0	1,600	—	-0.670	07/04/2008	06/25/2010
37	GeZhou	580025	Call	0	302	—	0.424	07/11/2008	12/31/2009
38	JiangTong	580026	Call	0	1,761	—	0.654	10/10/2008	09/21/2010
39	ChangHong	580027	Call	0	573	—	0.198	08/19/2009	08/11/2011
40	NanHang	580989	Put	1	1,400	12,832	-0.132	06/21/2007	06/13/2008
41	MaoTai	580990	Put	1	432	768	-1.141	05/30/2006	05/22/2007
42	HaiEr	580991	Put	1	607	767	-1.303	05/17/2006	05/09/2007
43	YaGe	580992	Put	1	635	754	-1.866	05/22/2006	05/14/2007
44	WanHua	580993	Put	1	85	191	-1.436	04/27/2006	04/19/2007
45	YuanShui	580994	Put	1	280	366	-0.289	04/19/2006	02/05/2007
46	BaoGang	580995	Put	1	715	834	-0.878	03/31/2006	03/23/2007
47	HuChang	580996	Put	1	568	593	-0.647	03/07/2006	02/27/2007
48	ZhaoHang	580997	Put	1	2,241	6,536	-1.969	03/02/2006	08/24/2007

This table reports, for each warrant traded in the period January 04, 2007 to October 16, 2009, its name, code, type, source, the number (in millions) of circulating shares offered at the beginning, the maximum number of circulating shares during its life span, moneyness ($\text{Money} = \log S_T / K$) for calls; $\text{Money} = -\log S_T / K$ for puts)

(a) Table 1: meta information for all warrants

(in minutes), gross DRR, net DRR, and DTO

Investor type	Percentiles					Mean	Skew.	Total
	1st	25th	Median	75th	99th			
A. Turnover value (million yuan)								
All	0.000	0.021	0.127	0.734	55.694	6.691	91.034	1,497.149
Institutions	0.001	0.377	2.892	15.143	796.104	41.453	5.851	10.197
Individuals	0.000	0.021	0.127	0.731	54.871	6.653	91.058	1,486.952
B. Gross profit (million yuan)								
All	-0.146	-0.003	-0.000	0.000	0.072	-0.003	61.561	-670.174
Institutions	-2.023	-0.028	-0.001	0.009	2.412	0.073	9.097	17.881
Individuals	-0.144	-0.003	-0.000	0.000	0.071	-0.003	63.053	-688.055
C. Net profit (million yuan)								
All	-0.176	-0.005	-0.001	0.000	0.052	-0.007	52.378	-1,462.944
Institutions	-2.123	-0.054	-0.001	0.006	2.307	0.039	8.675	9.580
Individuals	-0.174	-0.005	-0.001	0.000	0.051	-0.007	53.075	-1,472.524
D. Net profit per turnover (%)								
All	-48.82	-1.99	-0.61	0.00	5.57	-2.80	-253.37	—
Institutions	-59.24	-1.20	-0.24	0.44	14.56	-2.02	-6.62	—
Individuals	-48.76	-1.99	-0.61	0.00	5.57	-2.80	-253.28	—
E. Median of duration of trading cycle (minutes)								
All	2.85	30.38	79.91	244.61	15,422.99	759.15	11.59	—
Institutions	2.13	30.47	89.95	313.28	32,146.49	1,202.93	8.61	—
Individuals	2.85	30.38	79.89	244.55	15,397.71	758.67	11.59	—
F. Average gross DRR (bps)								
All	-1,682.01	-124.20	-29.59	28.39	611.62	-93.96	32.72	—
Institutions	-1,784.75	-104.74	-18.91	53.73	562.32	-50.79	-2.83	—
Individuals	-1,681.80	-124.24	-29.61	28.37	611.90	-94.01	32.72	—
G. Average net DRR (bps)								
All	-1,821.53	-184.14	-61.89	0.30	525.68	-147.08	28.82	—
Institutions	-1,793.38	-128.98	-29.76	38.06	548.16	-75.15	-2.80	—
Individuals	-1,821.72	-184.21	-61.91	0.26	525.66	-147.16	28.83	—
H. Average DTO (%)								
All	1.34	25.36	70.75	100.71	378.20	81.62	16.79	—
Institutions	1.11	24.42	65.29	101.20	354.62	84.37	12.16	—
Individuals	1.34	25.36	70.76	100.71	378.25	81.62	16.80	—

(b) Table 2: turnover, profits, DRR, and DTO

5.2 Table 3 + Table 4: Investor characteristics and luck versus skill

Read:

- Table 3 shows that individuals trade much more actively than institutions. The Individual coefficient in DTO regressions is about 70 percentage points.
- Women have higher gross and net DRRs and lower DTO than men in the individual-investor sample.
- Older and more experienced accounts perform better and trade less aggressively.
- Table 4 compares actual top/bottom investor performance with simulated performance.
- Top performers' actual mean-return statistics exceed simulated benchmarks; bottom performers' actual statistics are worse than the simulated distribution.

Economic message. The cross-section is not random noise. Demographics, experience, and trading intensity matter, and a small group of top performers shows evidence of skill rather than luck.

	A. Gross DRR		B. Net DRR		C. DTO	
Individual (dummy)	2.03 (0.05)	24.80 (0.54)	-44.67 (-1.12)	-27.78 (-0.62)	77.82*** (6.36)	69.80*** (5.48)
Skew×Individual	-1.37*** (-2.97)	-1.50** (-2.17)	-1.95*** (-4.16)	-1.78** (-2.56)	-0.23*** (-4.88)	-0.16** (-2.18)
May30×Individual	15.06 (0.38)	9.23 (0.24)	32.00 (0.82)	18.09 (0.49)	-32.43** (-2.35)	-30.95** (-2.18)
Age×Individual	0.87*** (8.46)	0.71*** (7.69)	1.57*** (13.76)	1.37*** (13.97)	0.40 (1.52)	0.41 (1.56)
Gender×Individual	12.70*** (5.13)	10.60*** (4.46)	20.64*** (8.00)	18.06*** (7.40)	-5.69 (-1.53)	-5.59 (-1.51)
AccountAge×Individual	2.28*** (6.00)	2.21*** (6.15)	3.06*** (7.59)	3.04*** (7.92)	-6.08*** (-11.28)	-6.05*** (-11.26)
Controls	No	Yes	No	Yes	No	Yes

This table presents results of panel regressions of investors' gross DRR (in basis points), net DRR (in basis points), and DTO (expressed as a percentage). The regressions include day fixed effects. Individual investors' demographic information, such as gender, age, and account age, is also included. "Gender" is a dummy variable that equals 1 for women and 0 for men. Age and account age are measured in years computed on January 1, 2010. Control variables include leverage, moneyness, delta, Vega, gamma, relative value, time-to-maturity, and price. Leverage is the option elasticity (i.e., the leverage implicit in the option) defined as the percentage change in the option price relative to the percentage change in the underlying. Volatility for a warrant on trading day t is defined as $(High_t - Low_t) / Close_{t-1}$, where $High_t$ is day high on t , Low_t is day low on t , and $Close_{t-1}$ is the closing price on $t-1$. Moneyness is defined as $\log(S/K)$ for calls, and as $-\log(S/K)$ for puts. Delta is defined as the absolute value of Black-Scholes delta for options. "May30" is a dummy variable equal to 1 on or after May 30, 2007, and equal to 0 otherwise. The relative value ("RelValue") is defined as the Black-Scholes value over the transaction price. Other variables are computed under the Black-Scholes model. Skewness is measured by percentile, because it is highly skewed. The relative value ("RelValue") is defined as the Black-Scholes value over the transaction price. Other variables are computed under the Black-Scholes model. Skewness is measured by percentile, because it is highly skewed. Skewness and controls are defined as value-weighted mean (across all trades) for each investor-day. t -statistics are based on robust standard errors clustered by investors and by date.

(a) Table 3: panel regressions of DRR and DTO

Pct	A. Best-performing investors				B. Worst underperforming investors			
	SimMean	SimSD	SimMax	Act	SimMean	SimSD	SimMin	Act
0.1	-4.70	1.00	-2.84	0.71	-2.90	0.23	-4.02	-9.69
1	-3.00	0.25	-2.20	0.76	-2.18	0.10	-2.56	-7.25
2	-2.54	0.17	-1.89	0.80	-1.93	0.07	-2.20	-6.16
3	-2.28	0.14	-1.73	0.82	-1.77	0.06	-2.00	-5.94
4	-2.09	0.12	-1.64	0.85	-1.65	0.06	-1.86	-5.72
5	-1.94	0.11	-1.48	0.87	-1.55	0.05	-1.77	-5.52
10	-1.47	0.08	-1.18	0.98	-1.22	0.04	-1.37	-5.04
20	-0.94	0.06	-0.69	1.15	-0.81	0.04	-0.98	-4.42
30	-0.59	0.06	-0.38	1.32	-0.50	0.04	-0.64	-3.98
40	-0.29	0.05	-0.09	1.46	-0.23	0.04	-0.37	-3.75
50	-0.03	0.05	0.16	1.64	0.02	0.04	-0.11	-3.49
60	0.23	0.05	0.41	1.91	0.28	0.04	0.14	-3.24
70	0.49	0.05	0.66	2.24	0.57	0.04	0.42	-3.01
80	0.79	0.05	1.01	2.78	0.91	0.04	0.75	-2.79
90	1.20	0.06	1.40	4.10	1.41	0.06	1.21	-2.57
95	1.52	0.07	1.81	6.12	1.84	0.07	1.56	-2.38
96	1.62	0.08	1.91	6.72	1.97	0.08	1.65	-2.34
97	1.73	0.09	2.05	7.55	2.13	0.09	1.79	-2.28
98	1.89	0.10	2.26	8.15	2.36	0.11	1.98	-2.22
99	2.14	0.13	2.69	9.71	2.73	0.15	2.26	-2.11
99.9	2.87	0.33	4.58	12.22	4.10	0.59	2.72	-1.95

The table reports t -values of the means at selected percentiles (Pct) of the distribution of t (mean) for actual (Act) net returns. Panel A reports results for the 682 best-performing investors, and panel B reports results for the 1,267 most underperforming investors. We identify best-performing investors as those who traded at least on 100 trading days and have Sharpe ratios no less than 1. We identify the most underperforming investors as those who traded at least on 100 trading days and have Sharpe ratios lower than -3. For each investor, we run 10,000 bootstrap simulations on their mean values adjusted daily returns that is, we subtract mean returns from the

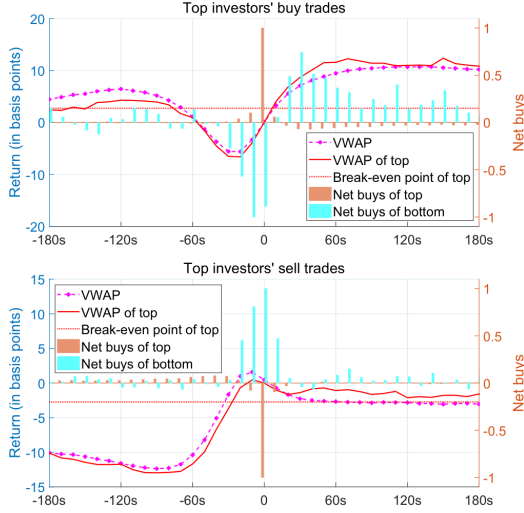
(b) Table 4: luck versus skill

5.3 Figure 1 + Table 5: Top traders and persistent performance

Read:

- Figure 1 shows that top investors' buys occur after short-run price declines, and their sells occur after short-run price increases.
- Bottom investors tend to take the other side, consistent with trend-chasing behavior.
- Table 5 groups investors by Sharpe ratios in a holdout period.
- The top 1% have high in-sample net DRR and high DTO, and they continue to perform better out of sample.
- The bottom 1% have strongly negative net DRR and continue to do poorly out of sample.

Economic message. Winners do not appear to be passive bubble riders. They trade actively and provide liquidity to investors who chase recent price movements.



(a) Figure 1: top investors' trades

Investor group	Sharpe ratio	net DRR	DTO	ValpTr	NoTrade	Turn Value	NoI
A. In sample (year 2007)							
Top 1%	6.00***	1.73***	730.33	314.76	28.04	24.47	49
1%10%	2.07***	1.10***	146.78	78.84	6.22	31.03	503
10%30%	0.41***	0.21	86.69	43.41	4.40	13.78	1,136
30%70%	-1.13***	-0.62**	81.37	40.89	4.20	19.56	2,275
70%90%	-2.63***	-1.30***	89.60	34.39	4.41	7.77	1,140
90%99%	-4.40***	-1.96***	133.58	27.08	5.23	2.94	494
Bottom 1%	-7.75***	-2.74***	260.65	24.53	8.14	0.45	55
Difference	13.75***	4.48***	469.68***	290.23***	19.90***	—	—
(t-stat)	(43.08)	(12.04)	(3.63)	(5.26)	(4.34)	—	—
B. Out of sample (years 2008–2009)							
Top 1%	3.51**	0.41***	505.98	290.96	23.90	24.35	49
1%10%	-1.09***	-0.31*	121.60	65.35	5.50	44.38	503
10%30%	-1.31***	-0.40**	68.48	30.63	3.43	12.02	1,136
30%70%	-1.43***	-0.46**	65.88	28.25	3.34	13.42	2,275
70%90%	-2.00***	-0.56***	63.59	22.28	3.18	3.92	1,140
90%99%	-3.46***	-0.99***	87.73	20.68	3.65	1.49	494
Bottom 1%	-3.93***	-1.12***	177.62	18.08	5.49	0.42	55
Difference	7.44***	1.54***	328.36*	272.88***	18.41***	—	—
(t-stat)	(5.36)	(7.52)	(1.88)	(3.97)	(2.62)	—	—

We group investors by their Sharpe ratios of net DRRs for warrant trading, and report the cross-sectional averages (within each group) of each investor's time-series averages of net DRR (expressed as a percentage), DTO (expressed as a percentage), value per trade (ValpTr, in thousands of yuan), and number of trades (NoTrade) per day with trading. The first column reports the cross-sectional averages of Sharpe ratios. The last two columns report aggregated turnover value (TurnValue, expressed as a percentage, w.r.t. total turnover value) and number of investors (NoI) in each group. We first identify investors who traded on at least 52 days in year 2007. A number of investors (6,441) satisfy this criterion. We then stratify these 6,441 investors by their Sharpe ratios of net DRRs for warrant trading. This table only reports summaries for those 5,652 investors who traded on at least

(b) Table 5: investor groups by Sharpe ratio

5.4 Figure 2 + Figure 3: Ex post performance and FF tests

Read:

- Figure 2 shows that among top investors, high-DTO groups have better ex post performance.
- The lowest DTO group is much more likely to suffer ex post losses.
- Figure 3 shows that many top investors pass a bootstrap-based FF test, and pass rates are highest for high-DTO groups.

Economic message. High turnover is not automatically bad. For the skilled subset, high DTO is associated with liquidity provision and persistence of profitability.

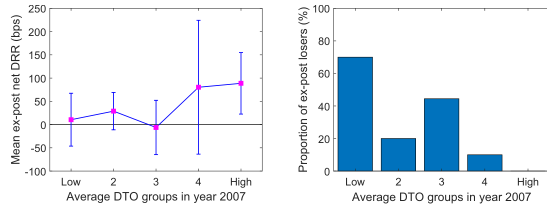


Figure 2
Top investors' ex post performance
Here, the top investors are the top 1% of investors identified in Table 5. The left panel shows cross-sectional

(a) Figure 2: top investors' ex post performance

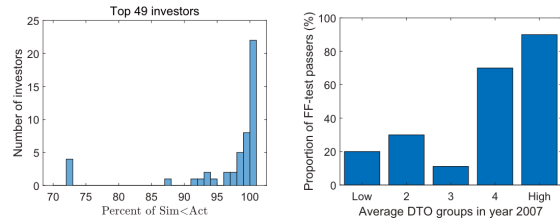


Figure 3
FF test for top investors

(b) Figure 3: FF tests for top investors

5.5 Table 6 + Table 7: Supertraders as de facto market makers

Read:

- Table 6 focuses on the top 1% and splits them by DTO.
- The highest-DTO group has in-sample DTO above 1,900%, high Sharpe ratios, short trading duration, frequent trading days, and a 100% winner rate.
- Out of sample, the high-DTO group continues to perform well and has a 100% winner rate.

- Table 7 shows that supertraders' large buys occur after negative returns, while their large sells occur after positive returns.

Economic message. The most successful investors are better described as market makers than directional speculators. They earn small profits per unit turnover but trade with enormous frequency.

Group	DTO	SharpeRatio	NetDRR	Prof/Turn	Duration	FracDay	Winner	NoI
<i>A. In sample (year 2007)</i>								
Low	115.95	5.62***	2.18***	19.42	2,794.00**	37.31	100.00	10
2	273.37	4.90***	1.10**	14.23***	255.00***	39.30	100.00	10
3	410.99	5.84***	2.01***	16.33***	226.61***	41.41	100.00	9
4	836.78	6.72***	1.31***	7.36***	107.65***	53.06	100.00	10
High	1,982.61	6.90***	2.08***	4.46***	93.00***	56.57	100.00	10
High-Low	1,866.66***	1.28	-0.10	-14.96	-2,701.00**	19.26**	0.00	
p-value	<.001	.127	.888	.700	.037	.016	1.000	
<i>B. Out of sample (years 2008–2009)</i>								
Low	115.53	-0.22	0.10	-14.29	2,345.95*	11.44	30.00	10
2	214.48	2.66*	0.29**	6.07	426.35**	17.18	80.00	10
3	316.18	7.88	-0.06	-0.51	344.72***	25.64	55.56	9
4	626.75	3.63***	0.80	3.79***	112.00***	21.31	90.00	10
High	1,237.99	4.03***	0.89***	3.83**	133.60**	51.77	100.00	10
High-Low	1,122.46***	4.25***	0.78***	18.12	-2,212.35	40.32***	70.00***	
p-value	<.001	.009	.009	.249	.109	.004	.001	

We group the top 1% of investors identified in Table 5 by their average DTO for warrant trading in 2007. For DTO (expressed as a percentage) and net DRR (expressed as a percentage), we report cross-sectional averages (within each group) of each investor's time-series averages. Sharpe ratio, profit per turnover (Prof/Turn, in basis points), median duration of trading cycles (Duration, in seconds), and the fraction of days with trading (FracDay, expressed as a percentage) are summary numbers for each investor that do not vary across days, and we report their respective cross-sectional averages. For each group, we also report the proportion of investors with positive net profits (Winner, expressed as a percentage). The last column reports the number of investors (NoI) in each group. In the last two rows of each panel, we report the differences between the high and low quintiles, as well as the corresponding *p*-values for testing the null of no difference. *p*-values for differences in DTO and NetDRR are based on cluster-robust standard errors accounting for auto- and cross-correlation. *p*-values for differences in Sharpe ratios, profit per turnover, and the fraction of days with trading are based on the two sample *t*-test. *p*-values for differences in proportions are based on the two-proportion *z*-test. The *p*-value for differences in medians is based on the Mood's median test. **p* < .1; ***p* < .05; ****p* < .01.

	-5min	-4	-3	-2	-1	<i>t</i> =0	1	2	3	4	5min
<i>A. Warrant's average return around STs' buy trades</i>											
Return	16.30*	27.07*	32.01***	19.77*	-42.20***	13.08	41.43***	-1.64	-5.36	13.34	-5.27
<i>t</i> -stat	(1.81)	(1.94)	(3.06)	(1.76)	(-3.55)	(1.20)	(4.26)	(-0.19)	(-0.62)	(0.97)	(-0.47)
Obs	299	299	300	300	300	302	302	302	302	301	300
<i>B. Warrant's average return around STs' sell trades</i>											
Return	15.64	8.88	-12.10	-7.49	53.59***	22.56**	-44.97***	2.34	15.14	-1.40	-6.40
<i>t</i> -stat	(1.15)	(0.81)	(-1.15)	(-0.68)	(5.81)	(2.19)	(-4.85)	(0.16)	(1.03)	(-0.14)	(-0.55)
Obs	315	316	316	318	320	320	320	320	319	318	317

This table reports the average returns (in basis points) around supertraders' large net buys and net sells in year 2007. Here, the supertraders (STs) are defined as the ten investors with the highest average DTO in year 2007 among the top-49 investors identified in Table 5. *t*=0 is the minute where the magnitude of STs' net buy (or sell) value exceeds one million yuan. Panel A reports results for net buying, and panel B reports results for net selling. *t*-statistics are based on robust standard errors clustered by warrant and by date. **p* < .1; ***p* < .05; ****p* < .01.

(a) Table 6: characteristics of the top 1%

(b) Table 7: supertrader event study

5.6 Table 8 + Figure 4: Bottom investors in OTM warrants

Read:

- Table 8 tracks trading volume, holdings, and returns during the last 10 days of OTM-expiring warrants.
- Top investors largely avoid these positions near expiration.
- Bottom-performing investors continue to trade and hold, and their net DRRs are negative.
- Figure 4 shows bottom investors increasing holdings as OTM warrants approach maturity, while top investors do not.

Economic message. Bottom investors behave as if warrants are lottery-like stocks. This is a direct channel through which naive traders lose money in the bubble.

Table 8 Trading volume, holding positions, and returns during the last 10 trading days of OTM expiring warrants											
Number of trading days to the last trading day										Differences	
Groups	(10)	(9)	(8)	(7)	(6)	(5)	(4)	(3)	(2)	(1)	(1)-(10)
<i>A. Trading volume</i>											
Top 0.1%	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
0.1%–1%	38.00***	40.69***	39.01***	43.43***	41.82***	37.61***	31.68***	42.20***	23.47***	7.61***	-30.38***
1%–10%	21.21***	32.97***	27.31***	16.68***	18.60***	22.60***	27.36***	22.70***	23.34***	27.30***	6.09
10%–90%	36.94***	22.39***	29.48***	30.69***	34.52***	36.07***	35.23***	31.43***	45.67***	57.26***	20.32*
Bottom 10%	3.86***	3.95***	4.18***	9.21***	5.06***	3.72**	5.73**	3.68***	7.52***	7.83***	3.97**
<i>B. Holding position</i>											
Top 0.1%	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
0.1%–1%	0.46	3.06	2.10*	4.15	7.54	4.99	0.01	0.10	0.00	0.00	-0.46
1%–10%	14.80**	18.92**	15.31*	3.64***	4.31***	10.32***	8.82***	11.77**	11.57**	8.55	-4.99
10%–90%	78.59***	69.52***	65.53***	81.51***	85.80***	80.59***	84.80***	80.68***	80.52***	75.80***	-2.91
Bottom 10%	6.74**	8.49***	17.06***	10.69**	2.35***	4.11***	6.36***	7.44**	7.91**	15.57***	8.83*
<i>C. Net DRR</i>											
Top 0.1%	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	—
0.1%–1%	-0.09	0.31	0.10	-0.28	0.30*	-0.22	-0.18	-0.10	-0.13	-0.34	—
1%–10%	-0.09	0.07	-0.48	-0.29**	-0.25	-0.43***	-0.89*	-1.49*	-1.16***	-0.84***	—
10%–90%	-0.20	0.30	-0.40	-0.25*	-0.23	-0.44***	-1.07*	-1.44**	-1.48***	-2.01***	—
Bottom 10%	-0.18	0.08	-0.41*	-0.12**	-0.24	-0.53***	-0.87*	-1.78***	-1.47***	-2.69***	—
Top-Bottom	0.18	-0.08	0.41*	0.12**	0.24	0.53***	0.87*	1.78***	1.47***	2.69***	—
N	8	8	8	8	8	8	7	7	7	7	—

Panels A and B of this table report the average (across warrants) proportion (expressed as a percentage) of daily trading volume and day-end holding positions contributed by different groups of investors during the last 10 trading days for OTM warrants, respectively. Panel C reports the cross-sectional averages of net DRRs on each value of the number of days to the last trading day. We only consider investors who traded on at least 52 days in year 2007 and at least 3 days after 2007. We stratify investors based on their Sharpe ratios or net DRRs for warrant trading in year 2007, and report summaries for the period from 2008 to 2009. Column "(1)" reports results for the last trading day, Column "(2)" reports results for the penultimate trading day, and so on and so forth. Column "(1)-(10)" reports the difference between Columns "(1)" and "(10)". Column "(1)-(5)" is defined analogously. The last row reports the sample size for each column. **p* < .1; ***p* < .05; ****p* < .01.

Table 8: trading volume, holdings, and returns in the last 10 trading days

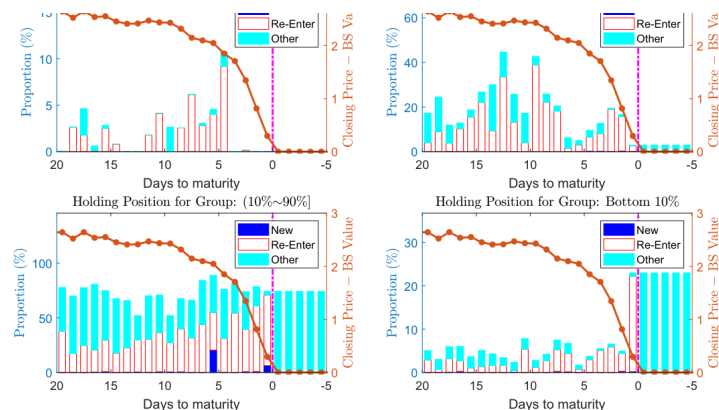


Figure 4

Day-end holding positions in OTM warrants during the last 20 trading days in the post-2007 period

This figure shows the proportions of aggregated day-end holding positions in all OTM (as per the moneyness on the last trading day) warrants for different performance groups (0.1%~1%, 1%~10%, 10%~90%, and bottom 10%). We stratify investors by their Sharpe ratios of net DRRs in year 2007 and follow the method used in Table 5. The top 0.1% of investors never trade these warrants in the post-2007 period. For each trading day, we split holding positions into three parts: the first (New) consists of positions held by new investors (investors that first trade the warrant on that day), the second (Re-enter) is the sum of net daily position changes by existing investors who increase positions on that day, and the third (Other) includes the remaining positions. In each plot, we also include the average overpricing (closing price minus the corresponding Black-Scholes estimate) on the right

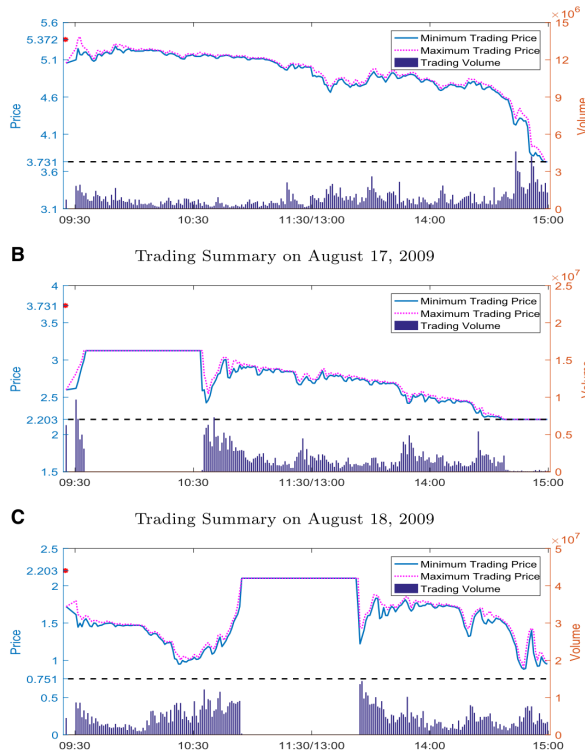
Figure 6: Figure 4: day-end positions in OTM warrants

5.7 Figure 5 + Figure 6: The 580018 zero-payoff episode and investor beliefs

Read:

- Figure 5 shows the final trading days of warrant 580018, including sharp price movements, trading suspensions, and trading volumes.
- The warrant eventually has a guaranteed zero payoff, yet active trading continues.
- Figure 6 shows an online forum where investors discuss suspended trading and possible resumption, revealing confusion and lottery-like beliefs.

Economic message. The episode is a vivid example of option-payoff misunderstanding. Some traders continued to treat a derivative with essentially no value as if it were an ordinary stock that might rebound.



(a) Figure 5: trading summary for warrant 580018

English translation of posts on the right hand side

Zhongyuan CWB1 will resume trading tomorrow

Slumped so much in past several days? Dang it! I will make a fortune! Why doesn't it resume trading? I will get married after the value (of my warrants) doubles! I exercised 100 shares. Will this warrant disappear forever?

Why is the trading (in warrants) suspended?

I will make a fortune: I bought 80,000 shares at 1.620, now it is at 19! [In fact, the strike price is 19]

The stock market has crashed. We are lucky since the warrant's trading is suspended.

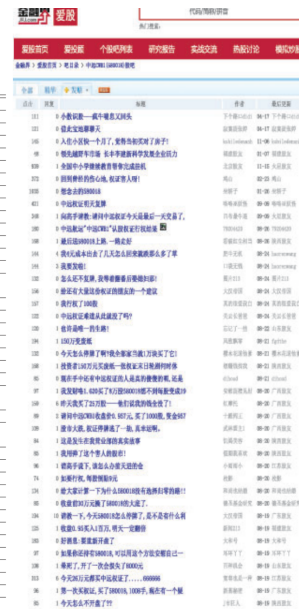
I bought 0.3 million yuan worth of warrants near market closing. Why is the trading suspended? Any good news? Bought 1 million at 0.95. Must be doubled tomorrow! Good news: [the warrant] will resume trading.

Invested 0.26 million in the warrant today awesome!

Why isn't the [warrant] market open today??

Figure 6 Screenshot of an online forum for warrant 580018

This figure shows a screenshot of a web page (<http://istock.jrj.com.cn/list/580018.p1.html>) for a forum on warrant 580018. The last trading day was August 18, 2009, and its maturity day was August 25, 2009. We translate some



(b) Figure 6: online forum screenshot

5.8 Table 9 + Figure 7: Zero-payoff warrants and order-book mechanics

Read:

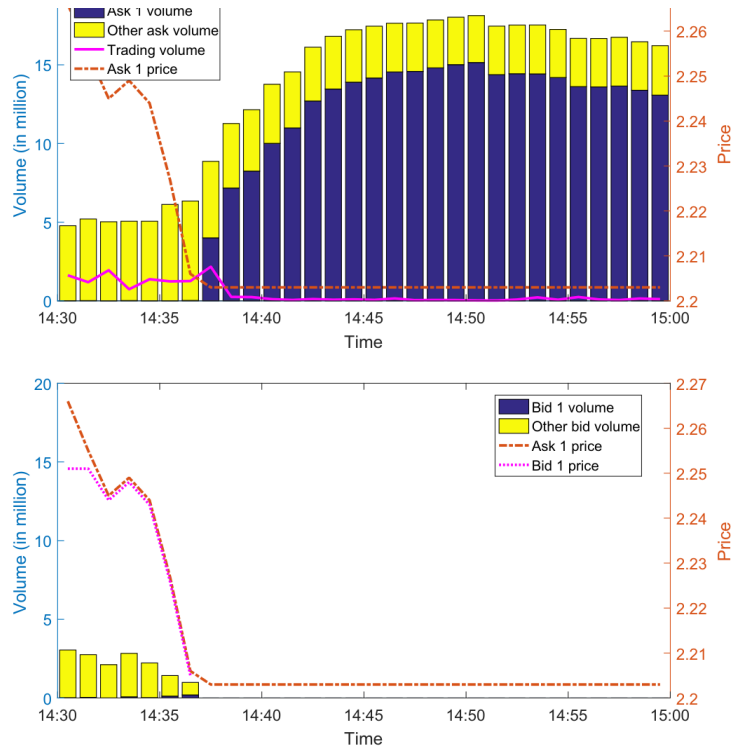
- Table 9 summarizes warrants with guaranteed zero payoff on the last trading day.
- Trading volume and holdings remain economically meaningful even when payoff is guaranteed to be zero.
- Figure 7 shows the order book and trading data for 580018 near the end, including limit-price dynamics and unavailable bid prices.

Economic message. The zero-payoff evidence is hard to reconcile with fully informed option trading. It points to confusion, gambling motives, and constraints embedded in market rules.

Code	Type	A. The first day with zero payoff							B. The last trading day						
		Date	High	Low	Close	Mean	Volume	HoValue	Fees	Date	High	Low	Close	Mean	Volume
580014	Call	10/19/2009	1.674	1.171	1.199	1.389	—	—	—	10/22/2009	0.545	0.011	0.016	0.408	—
580018	Call	08/18/2009	2.100	0.831	0.957	1.438	85.016	0.713	0.079	08/18/2009	—	—	—	—	—
580019	Call	02/24/2010	0.018	0.001	0.001	0.007	—	—	—	02/24/2010	0.550	0.024	0.024	0.313	533.604
580020	Call	02/25/2009	1.497	1.006	1.023	1.384	342.221	8.139	0.266	06/25/2010	0.016	0.001	0.001	0.009	—
580024	Call	06/24/2010	0.024	0.010	0.011	0.018	—	—	—	04/24/2007	0.320	0.008	0.010	0.137	125.940
038001	Put	04/17/2007	0.780	0.681	0.716	0.725	29.993	2.131	0.032	02/22/2008	0.150	0.010	0.010	0.078	526.993
038003	Put	02/18/2008	0.210	0.178	0.190	0.197	102.663	3.268	0.029	03/26/2008	0.080	0.004	0.004	0.049	511.248
038004	Put	03/10/2008	0.542	0.443	0.447	0.491	97.869	3.962	0.028	11/16/2007	0.107	0.010	0.010	0.066	919.168
038006	Put	11/07/2007	0.546	0.430	0.438	0.499	194.714	4.884	0.090	06/22/2007	1.000	0.100	0.107	0.598	147.052
038008	Put	06/15/2007	6.990	4.080	5.400	5.542	116.868	11.803	0.687	05/22/2007	0.041	0.002	0.003	0.017	308.426
580990	Put	05/15/2007	0.335	0.257	0.299	0.300	195.879	4.215	0.085	05/09/2007	0.110	0.001	0.001	0.042	349.772
580991	Put	04/24/2007	0.485	0.417	0.480	0.449	102.479	6.361	0.055	05/14/2007	0.059	0.002	0.002	0.022	386.160
580992	Put	04/23/2007	0.482	0.442	0.451	0.457	25.793	3.967	0.011	04/19/2007	0.441	0.067	0.087	0.178	164.527
580993	Put	04/19/2007	0.732	0.595	0.598	0.675	21.204	1.962	0.019	03/23/2007	0.270	0.005	0.006	0.104	421.288
580995	Put	03/21/2007	0.501	0.375	0.491	0.426	163.824	7.992	0.103	02/27/2007	0.442	0.012	0.015	0.188	226.641
580996	Put	02/16/2007	0.527	0.441	0.500	0.478	96.682	4.224	0.078	08/24/2007	0.176	0.001	0.002	0.057	2,953.652
580997	Put	08/08/2007	0.345	0.298	0.303	0.316	663.677	67.858	0.235	—	—	—	—	—	—
Total							2,238.882	131.479	1.797						7,309.716

This table reports the trading summaries for 17 warrants that have guaranteed zero payoff on at least one trading day of their life span. Panel A reports statistics for the first day that the warrant has a guaranteed zero payoff at market close, and panel B reports results for the last trading day. We report the day high, day low, closing price, volume weighted mean price, trading volume (in million shares), holding value (HoValue, hold position times the best bid at market closing; in million yuan), and commission fees (in million yuan). Here, the volume, value and fees are computed from our data set, rather than from the whole market. The maximum possible payoff at market close is computed according to the price limit rule in the stock market: for call warrants we assume the underlying stock increases by 10% per trading day, while for put warrants we assume the underlying stock decreases by 10% per trading day.

Table 9: warrants with guaranteed zero payoff on the last trading day



re 7
er book and trading data for warrant 580018

Figure 8: Figure 7: order book and trading data for warrant 580018

5.9 Table 10 + Figure 8: Price limits and regulatory failure

Read:

- Table 10 counts days when the lower price limit exceeds Black-Scholes values or maximum possible payoffs.
- Several warrants have many days in which the price limit is above the theoretical value; some have price limits above maximum possible payoff.
- Figure 8 shows that top investors reduce or avoid positions around problematic lower limits, while bottom investors continue to hold.

Economic message. Regulatory price limits can fail when they are calibrated without understanding fundamental values and investor behavior. In this market, limits sometimes protected overpricing rather than prevented crashes.

Code	Type	N_{BS1}	N_{BS2}	N_{Max}
031004	Call	6	5	0
031005	Call	13	10	0
031006	Call	2	0	0
580010	Call	5	4	0
580012	Call	7	5	0
580013	Call	14	14	0
580014	Call	15	15	4
580015	Call	11	10	0
580017	Call	8	4	0
580018	Call	16	16	2
580019	Call	1	1	0
580020	Call	16	16	4
580021	Call	14	13	0
580023	Call	11	2	0
580025	Call	1	1	0
038008	Put	1	1	0
580989	Put	2	2	0

The table reports the number of trading days (during the last 20 trading days of each warrant) on which warrants' lower price limit for trading is higher than their day-end theoretical prices. Three proxies for theoretical prices are used here. The column N_{BS1} uses the Black-Scholes prices with volatility parameter set as the 60-day trailing historical volatility σ_{60} of underlying asset; the column N_{BS2} uses the Black-Scholes prices with σ_{60} as volatility parameter; and the column N_{Max} uses the warrant's maximum

(a) Table 10: theoretical prices and price limits

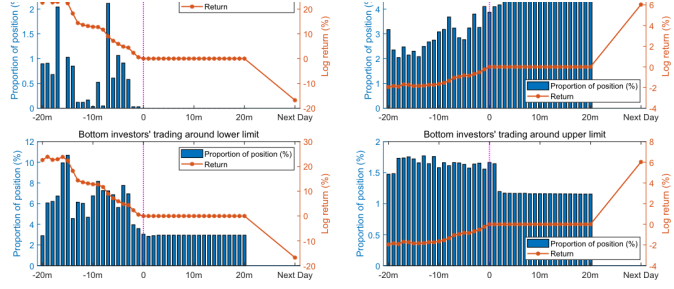


Figure 8

Proportions of aggregated holding positions of different classes of investors around price limits hits

Time 0 is the last time (in that day) that warrant price hits the associated price limits. In each minute, we compute the proportions of aggregated holding positions for each group of investors. For each warrant, we calculate intraday returns as log returns relative to the price limit they hit. The average return of the opening price on the next trading day relative to the hit price limit on the previous trading day is also reported. All 29 cases of upper limit hits are included in our analysis. For lower limit hits, we consider all four cases in which the lower price

(b) Figure 8: positions around price limits

6 Short summary

- The paper studies the Chinese warrants bubble using proprietary trader-level brokerage data.
- The active sample includes 223,745 investors and 48 warrants over 678 trading days.
- The market has transparent fundamental values because the securities are derivatives, making the bubble easier to measure than many equity bubbles.
- Investors on average lose money, especially after transaction costs.
- Individuals trade much more actively than institutions and lose more in aggregate.
- A small group of investors has persistent outperformance.
- The best-performing investors are not simply lucky; bootstrap tests and out-of-sample performance support skill.
- The skilled subset trades extremely actively, has short trading cycles, and buys after drops and sells after rises.
- This behavior resembles market making.
- Bottom-performing investors trend-chase and increase holdings in out-of-the-money warrants near expiration.
- Some traders continue to hold or trade warrants with guaranteed zero payoff.
- Regulators' price limits sometimes failed to bracket Black-Scholes values and maximum payoffs.
- The evidence shows how wealth is transferred from naive investors to sophisticated liquidity providers during a derivative bubble.

7 Conclusion

7.1 Core conclusion

Li, Subrahmanyam, and Yang show that bubbles are not only about aggregate overpricing. They are also about who trades with whom. In the Chinese warrants bubble, a small group of skilled traders earned persistent profits by acting as de facto market makers, while many unsophisticated traders lost money through trend chasing, lottery-like trading, and misunderstanding of option payoffs.

7.2 Economic intuition

The intuition is that derivative bubbles combine valuation difficulty, market frictions, and heterogeneous investor sophistication. The warrant's fundamental value may be clear to a sophisticated trader, but naive traders may treat the warrant like a stock or lottery ticket. Skilled traders can profit by providing liquidity to these traders, especially when shorting and market structure prevent easy arbitrage.

7.3 Takeaway

The paper's key lesson is that regulatory tools must account for both fundamentals and investor behavior. Price limits that ignore option values can preserve overpricing and leave naive traders exposed. In bubbles, the identity and behavior of traders matter as much as the price path itself.